PINCHED JORDAN CURVES, CANNON-THURSTON MAPS, AND THEIR DECOMPOSITIONS

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Abstract. TODO

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1. Introduction

Definition 1.1. A *pinched Jordan curve* is a map $\varphi_\infty : S^1 \to S^2$ and a choice of pseudo-isotopy φ_s, $s \in [0, \infty]$ landing on φ_∞. To avoid unnecessary quotients, we require that each preimage $\varphi^{-1}(p)$ is totally disconnected. ♮

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2. Definitions and Preliminaries

In this section, we give the topological background required for the rest of the paper. We cover Hausdorff convergence, upper semi-continuous decompositions, and locally flat isotopies.

2.1. Convergence of Sets.

2.1.1. *Hausdorff Convergence.* Let X be a metric space. Writing $N_\varepsilon(A) = \bigcup_{a \in A} B_\varepsilon(a)$, we may endow $\mathcal{P}(X)$, the power set of X, with the *Hausdorff distance*:

$$d_H(A, B) = \inf\{r \mid A \subset N_r(B) \text{ and } B \subset N_r(A)\}$$

One need not work very hard to construct examples showing that both inclusions are needed for d_H to be symmetric. When X is a compact metric space, then d_H gives a metric on the closed subsets of X—d_H must be restricted the set of closed subsets of X rather that

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all of $\mathcal{P}(X)$ as $d_H(A, \overline{A}) = 0$ for all sets A. The Hausdorff distance sees two sets as close if they they are everywhere point-wise close. Indeed, we will often establish that two sets are Hausdorff-close via the following characterization:

$$d_H(A, B) < \varepsilon \Leftrightarrow A \subset N_\varepsilon(B) \text{ and } B \subset N_\varepsilon(A)$$

We write $\lim_H(A_n) = A$ or $A_n \xrightarrow{HD} A$ if A is closed and $d_H(A_n, A) \to 0$ as $n \to \infty$.

Lemma 2.1. *Hausdorff convergence respects uniformly continuous maps and uniformly convergent sequences. Let $f_k : X \to Y$ be a sequence of maps between compact metric spaces such that $f_k \to f_\infty$ uniformly. If $A_k \xrightarrow{HD} A$ then $f_k(A_k) \xrightarrow{HD} f_\infty(A)$.*

Proof. Fix $\varepsilon > 0$. As $f_k \to f_\infty$ is uniformly convergent, Arzelà–Ascoli implies that the family $\{f_k\}_{k \in \mathbb{Z}^+ \cup \{\infty\}}$ is uniformly equicontinuous. As such, there is $\delta > 0$ such that if $d_X(x, y) < \delta$ then $d_Y(f_k(x), f_k(y)) < \varepsilon$ for any $k \in \mathbb{Z}^+ \cup \{\infty\}$. Further, there is N_1 such that if $k > N_1$, then $d_Y(f_k(x), f_\infty(x)) < \varepsilon$. As $A_k \xrightarrow{HD} A$, there is some $N_2 > 0$ such that if $k > N_2$, then $d_{HD}(A_k, A) < \delta$.

Now suppose that $k > \max\{N_1, N_2\}$. Pick $y \in f(A)$ and $x \in f^{-1}(y)$. As $d_{HD}(A_k, A) < \varepsilon$, there is some $x_k \in A_k$ such that $d_X(x, x_k) < \delta$. Then

$$d_Y(y, A_k) \leq d_Y(y, f_k(x_k)) \leq d_Y(f(x), f_k(x)) + d_Y(f_k(x), f_k(x_k)) \leq 2\varepsilon,$$

hence $f(A) \subset N_{2\varepsilon}(f_k(A_k))$.

Now suppose that $y \in f_k(A_k)$. Select $x \in f_k^{-1}(y)$. As $x \in A_k$ there is some $x' \in A$ such that $d_X(x, x') < \delta$. Then

$$d_Y(y, f(A)) \leq d_Y(f_k(x), f(x')) \leq d_Y(f_k(x), f_k(x')) + d_Y(f_k(x'), f(x')) \leq 2\varepsilon.$$

We conclude that $f_k(A_k) \subset N_{2\varepsilon}(f(A))$. We have shown that if k is sufficiently large, then $d_{HD}(f_k(A_k), f(A)) < 2\varepsilon$. □

We will occasionally use a slightly different formulation for convergence. Suppose that X is a compact metric space and let X_n be a sequence of closed subsets of X. We define the limits inferior and superior as follows:[1]

$$\liminf X_n = \{x \in X \mid \text{ Every neighborhood of } x \text{ intersects all but finitely many } X_n\}$$
$$\limsup X_n = \{x \in X \mid \text{ Every neighborhood of } x \text{ intersects infinitely many } X_n\}$$

Equivalently, $x \in \liminf X_n$ if one can find a sequence $x_n \in X_n$ such that $x_n \to x$ and $x \in \limsup X_n$ if one can find a subsequence n_k and a sequence $x_k \in X_{n_k}$ such that $x_k \to x$. It is immediate that $\liminf X_n \subset \limsup X_n$. For any sequence of sets X_n, $\limsup X_n$ and $\liminf X_n$ are closed. Indeed, if $x_n \in \liminf X_n$ is a sequence converging to some $x \in X$ then any neighborhood U of x is also a neighborhood of x_n for some n. As $x_n \in \liminf X_n$, all but finitely many of the X_n meet U, hence $x \in \liminf X_n$—so $\liminf X_n$ is closed. The same argument applies to $\limsup X_n$.

Lemma 2.2. *Suppose that X_n is a sequence of closed sets in a compact metric space X. If $Y = \liminf X_n = \limsup X_n$, then $X_n \xrightarrow{HD} Y$.*

[1]The reader may recognize these as *Kuratowski* convergence, a formulation for taking limits of sets in general topological spaces. In a compact metric space these formulations agree so we do not make the distinction. See [Hocking Young Sec ??].

Proof. Fix $\varepsilon > 0$. First, we will establish that if n is sufficently large that $X_n \subset N_\varepsilon(Y)$. Suppose to the contrary, that is, there is a subsequence n_k such that for all k $X_{N_k} \setminus Y \neq \varnothing$; select an $x_k \in X_{N_k} \setminus Y$ for each k. After passing to a subsequence, we may assume that $x_k \to x$. Each x_k satisfies $d(x_k, Y) > \varepsilon$, so $d(x, Y) \geq \varepsilon$, but $x \in \limsup X_n = Y$, a contradiction. Thus, for all but finitely many n, $X_n \subset N_\varepsilon(Y)$.

Now we will establish that for large n, $Y \subset N_\varepsilon(X_n)$. Supposing otherwise, we obtain a subsequence n_k such that $Y \setminus N_\varepsilon(X_{n_k})$ is nonempty. For each k select some $x_k \in Y \setminus N_\varepsilon(X_{n_k})$; up to subsequence $x_k \to x \in Y$. We now claim that $N_{\varepsilon/2} \cap X_{n_k}$ is empty for all large k. Indeed if k is large enough that $d(x_k, x) < \varepsilon/2$, then

$$d(x_k, X_{n_k}) \leq d(x_k, x) + d(x, X_{n_k}) \leq \varepsilon.$$

This contradicts the fact that $x_k \notin N_\varepsilon(X_{n_k})$ so we conclude that for large k $N_{\varepsilon/2}(x) \cap X_{n_k} = \varnothing$. On the other hand, $x \in \liminf X_n$ so $N_{\varepsilon/2}(x)$ must contain all but finitely many of the X_n. This contradiction establishes that, for large n, $Y \subset N_\varepsilon(X_n)$. Thus, $X_n \xrightarrow{HD} Y$. □

We end this section with a number of useful facts concerning Hausdorff convergence.

Proposition 2.3 ([HY88], Theorem 2-101)**.** *If X_n is a sequence of connected sets in a compact Hausdorff space X and if $\liminf X_n$ is nonempty, then $\limsup X_n$ is connected.*

Corollary 2.4. *Suppose $X_n \to X$ is a convergent sequence in a compact Hausdorff space. If each X_n is connected, so is X.*

Proposition 2.5 ([HY88], Theorem 2-102)**.** *If M is a compact metric space, then every sequence of subsets of M has a Hausdorff convergent subsequence.*

Fact 2.6. *If $A_n \xrightarrow{HD} A$, then* $\operatorname{diam} A_n \to \operatorname{diam} A$

Fact 2.7. $\operatorname{diam}(A \cup B) \leq \operatorname{diam} A + \operatorname{diam} B + d(A, B)$, *where* $d(A, B) = \inf\{d(a, b) \mid a \in A, b \in B\}$.

2.2. **Decompositions.** For a space X, a *decomposition* of X, $\mathcal{D}$, is a partition of X into closed subsets. That is, we can write $X = \bigsqcup_{D \in \mathcal{D}} D$ and each $D \subset X$ is closed. For $x \in X$, we will write D_x for the unique decomposition element containing x. This gives a set function $\pi : X \to \mathcal{D}$ given by $x \mapsto D_x$. We may endow $\mathcal{D}$ with topology by declaring that $\mathcal{A} \subset \mathcal{D}$ is open when $\pi^{-1}(\mathcal{A})$ is open in X. This is precisely the quotient topology inherited from X—so π is continuous. In light of this, by a slight abuse of notation we will usually use $\mathcal{D}$ to represent both subsets of X and for the quotient space $X/\mathcal{D}$. If we wish to stress the fact that we are treating $\mathcal{D}$ as a space on its own, we will refer to it as a *decomposition space*.

Given any other map $f : X \to Y$ which is constant on $\mathcal{D}$, we obtain a unique $g : \mathcal{D} \to Y$ such that $f = g \circ \pi$. We say that a set $W \subset X$ is *$\mathcal{D}$-saturated* if W can be written as a union of decomposition elements. If $D \in \mathcal{D}$ contains more than one point, we say that D is a *non-degenerate* element of $\mathcal{D}$. Otherwise D is called a *degenerate* decomposition element. We refer the curious reader to [Dav] for a full treatment of the theory of decompositions of manifolds and [Kal21] for a slightly more general treatment.

Definition 2.8. A decomposition $\mathcal{D}$ of a space X is *upper semi-continuous* if, given any open set $U \subset X$ and some $D \subset U$, there is an open set $V \subset U$ containing D such that every decomposition element intersecting V is contained in U. ♮

Upper semi-continuous decompositions are meant to generalize point-preimages of a surjective map $f : X \to Y$ from a compact space to a Hausdorff space one. We create the

decomposition $\mathcal{D}$ by dividing X into point-preimages:

$$\mathcal{D} = \{f^{-1}(y) \mid y \in Y\} \setminus \varnothing.$$

$\mathcal{D}$ is then upper semicontinuous. It is a classic result that Y and $\mathcal{D}$ are therefore homeomorphic. While the point-preimage example is the model for upper semi-continuous decompositions, there are a number of other useful characterizations coming from a combination of [Kal21, Lemma 2.7] and [FL25, Lemma 2.8]

Proposition 2.9. *Let $\mathcal{D}$ be a decomposition of a compact metric space X. The following are equivalent:*

(1) $\mathcal{D}$ is upper semi-continuous
(2) For every closed $K \subset X$, then set $\bigcup\{D \in \mathcal{D} \mid D \cap K \neq \varnothing\}$ is closed in X.
(3) Let A_i be a sequence of decomposition elements with $\lim A_i = A$. Then there is a $D \in \mathcal{D}$ with $A \subset D$.
(4) The quotient $\pi : X \to \mathcal{D}$ is closed.
(5) As a decomposition space, $\mathcal{D}$ is Hausdorff.

2.3. Embeddings and Isotopy Extension.

Definition 2.10. A pseudo-isotopy is a continuous family of maps $f_t : X \to Y$, $t \in [0,1]$ such that for all $t < 1$, f_t is an embedding. ♮

The map $f_1 : X \to Y$ may or may not be an embedding. If $J \subset [0,1)$ is a closed interval, then the collection $f_t, t \in J$ is an *isotopy*.

Definition 2.11. An embedding of topological manifolds $N^n \subset M^m$ is called *locally flat* if for every $p \in N$, there is a chart $\phi_p : \mathbb{R}^m \to M$ centered at p such that $\mathbb{R}^{m-n} \subset \mathbb{R}^m$ is mapped to N. Embeddings which are not locally flat are called *wild* embeddings. ♮

Locally flat embeddings serve as the generalization of smoothly embedded submanifolds; wild embeddings can introduce pathologies. See, for example, [CITE: Rolfsen? Other?]. An embedding of spheres $S^n \hookrightarrow S^m$ is called *flat* if S^n be realized as the image of the equatorial n-sphere under a homeomorphism $S^m \to S^m$. The famous Schoenflies Theorem [CITE: Hocking Young? Earlier?] states that every S^1 embedded in S^2 is flat.

Definition 2.12 ([EK71])**.** An isotopy of manifolds $f_t : N^n \to M^m$, $t \in I$, is *locally flat* if for every $(p,t) \in N \times I$, there is a neighborhood $[t_0, t_1]$ of $t \in I$ and level-preserving embeddings $\alpha : \mathbb{R}^n \times [t_0,t_1] \to N \times I$ and $\beta : \mathbb{R}^n \times \mathbb{R}^{m-n} \times [t_0,t_1] \to M \times I$ onto neighborhoods of (p,t) and $(f_t(p),t)$ respectively such that the following diagram commutes.

$$\begin{array}{ccc} \mathbb{R}^n \times 0 \times [t_0,t_1] & \hookrightarrow & \mathbb{R}^n \times \mathbb{R}^{m-n} \times [t_0,t_1] \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} \\ N \times I & \xrightarrow[(p,t)\mapsto(f_t(p),t)]{} & M \times I \end{array}$$

In particular, this means that if U is a sufficiently small locally flat chart around $f_t(p)$ for some $p \in N$, then the isotopy locally extends to $F : U \times [t_0,t_1] \to M$ such that for all $t' \in [t_0,t_1]$, $F_{t'}(U)$ is a locally flat chart for $f_{t'}(p)$. ♮

As long as N is a locally flat proper submanifolds, a locally flat isotopy may be viewed as a level preserving map $N \times I \to M \times I, (p,t) \mapsto (f_t(p),t)$ which locally flat in the usually sense.

Definition 2.13. An isotopy $f_t : X \to Y$, $t \in I$, is *realized by an ambient isotopy* if there is an isotopy $F_t : Y \to Y$, $t \in I$, which cover the f_t in the sense that $F_1 = \mathbb{1}_Y$ and for all $t \in I$, $f_t = F_t f_0$. ♮

Theorem 2.14 ([EK71], Cor. 1.4)**.** *Suppose $f_t : N \to M$, $t \in I$ is a locally flat proper isotopy of a compact manifold N into a manifold M. Then f_t is realized by an ambient isotopy of M.*

In the context of Schoenflies theorem, one wonders which *isotopies* $f_t : S^n \subset S^m$ are locally flat. Work by Oversteegen and Tymchatyn [OT08] answers this in the case where $n = 1$ and $m = 2$:

Theorem 2.15 ([OT08], Thm. 7.2)**.** *Suppose that $Z \subset \mathbb{R}^2$ is compact and connected and that $f_t : Z \to S^2$ is an isotopy such that $f_0 = \mathbb{1}_Z$. Then there exists an ambient isotopy $F_t : S^2 \to S^2$ extending f_t.*

3. Pinched Jordan Curves and Inner Diameter Distance

We consider the class of curves $\varphi : S^1 \to S^2$ which are realized by a pseudo-isotopy:

Definition 3.1. A *pinched Jordan curve* is a map $\varphi_\infty : S^1 \to S^2$ and a choice of pseudo-isotopy φ_s, $s \in [0, \infty]$, landing on φ_∞. To avoid unnecessary quotients, we require each preimage $\varphi^{-1}(p)$ is totally disconnected. ♮

Remark 3.2. Given a pseudo-isotopy $\phi : S^1 \times I \to S^2$ where $\phi_\infty(S^1)$ does not have totally disconnected points preimages, we may pass to a quotient of S^1 to obtain a new map $\hat{\phi}_\infty$ which does not have totally disconnected preimages. Any arcs collapsed by ϕ_∞ must get uniformly small under the ϕ_s, so we may modify each ϕ_s to produce empeddings $\hat{\phi}_s$ such that $\hat{\phi}_s \to \hat{\phi}_\infty$. ♮

Remark 3.3. While our definition requires a *specific* pseudo-isotopy, it is a consequence of Theorem 6.1 that our results depend on an isotopy class of pseudo-isotopies landing on φ_∞. See Section 6. ♮

Notation. We fix the following notation which will be used throughout: For $s \in [0, \infty]$, let $C_s = \varphi_s(S^1)$. For $s < \infty$, C_s is a Jordan curve; thus, given some fixed orientation on S^1, C_s divides S^2 into two disks. Let $\mathring{C}_s^+$ and $\overline{C_s^+}$ denote the open and closed disks to the left of C_s. Likewise, we write $\mathring{C}_s^-$, and $\overline{C_s^-}$ for the open and closed disks to the right of C_s. ♮

Definition 3.4. Suppose φ_s, $s \in [0, \infty]$, be a pinched Jordan curve and let $p \in S^2 \setminus C_\infty$. We say that p is *to the left of* (resp. *to the right of*) C_∞ if $p \in \mathring{C}_s^+$ (resp. $p \in \mathring{C}_s^-$) for all sufficiently large s. ♮

Lemma 3.5. *Suppose φ_s, $s \in [0, \infty]$, is a pinched Jordan curve. Then*

1. *Every $p \in S^2 \setminus C_\infty$ either to the left or right of C_∞.*
2. *Every component U of $S^2 \setminus C_\infty$ consists entirely of points to the left or the right of C_∞.*

Proof. First, observe that p cannot be to the left *and* to the right of C_∞ as $\mathring{C}_s^+ \cap \mathring{C}_s^- = \varnothing$ for all s. Suppose, therefore, that $p \in S^2 \setminus C_\infty$ and that p is not to the right of C_∞. For sufficiently large s, $d_{S^2}(p, \overline{C_s^-}) > 0$; hence $p \in S^2 \setminus \overline{C_s^-} = \mathring{C}_s^+$—so p is to the left of C_∞.

Now let U be a component of $S^2 \setminus C_\infty$. Seeking a contradiction, suppose that $p, q \in U$ but p is to the left of C_∞ while q is to the right. Let α be an arc in U from p to q. As p is

to the left of C_∞, there is an open set $V_p \subset U$ containing p such that $V_p \subset \mathring{C}_s^+$ for large s. Similarly there is an open V_q containing q such that $V_q \subset \mathring{C}_s^-$ for large s. It follows that for sufficiently large s, C_s separates p and q—hence $C_s \cap \alpha$ is nonempty. But then $\alpha \cap C_\infty$ is also nonempty, but $\alpha \subset U$, a contradiction. □

Definition 3.6. We say that a pinched Jordan curve C_∞ has *no left side* if every complementary region is to the right of C_∞ and similarly define pinched Jordan curves with no right side. ♮

The following definition is inspired by [Cal07, Construction 10.11] and by section 10.4.1 more broadly.

Definition 3.7. Suppose φ_s, $s \in [0, \infty]$ is a pinched Jordan curve. Given $x, y \in S^1$ and $s < \infty$, define the *s-inner diameter* of x and y to be

$$d_s^\pm(x, y) = \inf\{\operatorname{diam} X \mid X \subset \overline{C_s^\pm} \text{ is connected, closed, and } \varphi_s(x), \varphi_s(y) \in X\}$$

The sets X above are called *s-stems* between x and y (or just a *stem* when x, y, and s are understood). ♮

Remark 3.8. As discussed in the introduction, we are mainly interested in pinched Jordan curves where $\varphi_\infty : S^1 \to S^2$ is surjective—meaning there is no left or right side. Since "left" and "right" are arbitrary, our generally arguments assume that φ has no left side and leave the right side as a symmetric argument. As such, for the rest of this section we only consider d_s^+. ♮

Definition 3.9. Suppose $\varphi_s, s \in [0, \infty]$, is a pinched Jordan curve and X is an s-stem for $x, y \in S^1$. We say that X is *taut* if $\operatorname{diam} X = d_s^+(x, y)$. We say that X is ε-efficient if X is a proper arc in $\overline{C_s^+}$ and $\operatorname{diam} X \leq d_s^+(x, y) + \varepsilon$. ♮

Lemma 3.10 (Existence of Taut and Efficent Stems)**.** *Suppose $\varphi_s, s \in [0, \infty]$ is a pinched Jordan curve. Given $x, y \in S^1$ and $s < \infty$:*

(1) There is an s-stem X such that $\operatorname{diam} X = d_s^+(x, y)$
(2) Given $\varepsilon > 0$ and any s-stem X for x and y, there is a proper arc $\gamma \subset \overline{C_s^+}$ which is also an s-stem for x and y such that $\operatorname{diam} \gamma \leq \operatorname{diam} X + \varepsilon$.

Proof.

(1) Let X_n be a sequence of closed, connected sets containing $\varphi_s(x)$ and $\varphi_s(y)$ such that $\operatorname{diam} X_n \to d_s^+(x, y)$. Passing to a subsequence if necessary, assume $X_n \xrightarrow{HD} X$. As $\varphi_s(x), \varphi_s(y) \in X_n$, for all n, $\varphi_s(x), \varphi_s(y) \in X$. By Corollary 2.4, X is connected. Finally, $d_s^+(x, y) = \lim_n \operatorname{diam} X_n = \operatorname{diam} \lim_H X_n = \operatorname{diam} X$, as required.
(2) Consider $N_\varepsilon(X)$, the ε-neighborhood of X in $\overline{C_s^+}$. $N_\varepsilon(X)$ is path connected, contains $\varphi_s(x)$ and $\varphi_s(y)$, and satisfies $\operatorname{diam} N_\varepsilon(X) \leq \operatorname{diam} X + \varepsilon$—thus, we select some proper arc $\gamma \subset N_\varepsilon(X) \subset \overline{C_s^+}$ containing $\varphi_s(x)$ and $\varphi_s(y)$.

□

Remark 3.11. In general, a stem cannot be both taut and ε-efficient. [Fab99] shows that for *planar* disks most taut stems contain segments of the boundary, but we note that their proof does directly not extend to S^2. ♮

Proposition 3.12. *For each $s < \infty$, d_s^+ is a metric on S^1.*

Proof. Symmetry and positive definiteness follow directly from definition and the fact that each φ_s is an embedding. For the triangle inequality, let X be a taut stem for $x, y \in S^1$ and let Y be a taut stem for $y, z \in S^1$. Then $X \cup Y$ is an s-stem for x and z, hence

$$d_s^+(x,z) \leq \operatorname{diam}(X \cup Y) \leq \operatorname{diam} X + \operatorname{diam} Y = d_s^+(x,y) + d_s^+(y,z),$$

where the final inequality follows from the fact that X and Y intersect at $\varphi_s(y)$.† □

lk: And an implicit use of Fact 2.7. Call this out directly?

3.1. **Curve Pushing and Continuity of Inner Distance.** While we cannot say much about C_∞ for a generic pinched Jordan curve, we may still define $d_\infty^\pm$ as the limit of the $d_s^\pm$. The goal of this section is two-fold: first, we wish to show that $\lim_s d_s^\pm(x,y)$ exists for any pair $x, y \in S^1$. Second, we use this fact to show $d_s^\pm$ is continuous as a function $[0,\infty] \times S^1 \times S^1 \to \mathbb{R}$.

Given α_s, an efficient s-stem for x and y and another $s' \in [0,\infty)$ such that φ_s and $\varphi_{s'}$ are ε-close as maps, we produce an s'-stem $X_{s'}$ such that the diameters of α_s and $X_{s'}$ are $O(\varepsilon)$-close. One would expect that most of α_s is still to the left of $C_{s'}$, so we would like $X_{s'}$ to contain $\alpha_s \cap \overline{C_{s'}^+}$ whenever possible. In fact, our construction produces a stem $X_{s'} \subset \varphi_{s'}(S^1) \cup (\alpha_s \cap \overline{C_{s'}^+})$. To achieve this, whenever the isotopy intersects α_s, we "push" those sections of α_s along with the isotopy.

Lemma 3.13. *Suppose $x_n \to x \in S^1$ and $s_n \to \infty$. Then $d_{s_n}(x_n, x) \to 0$.*

Proof. Let I denote a small interval around $x \in S^1$. Passing to the tail of the sequence, we any assume that $x_n \in I$ for all n. Let I_n denote the component of $S^1 \setminus \{x, x_n\}$ contained in I. As $\varphi_{s_n}(I_n)$ is an s_n-stem for x and y we know $d_{s_n}^\pm(x,y) \leq \operatorname{diam} \varphi_{s_n}(I_n)$. Further, $I_n \xrightarrow{HD} \{x\}$, hence $\varphi_{s_n}(I_n) \xrightarrow{HD} \{\varphi_\infty(x)\}$ by Lemma 2.1. Taking diameters, $d_{s_n}^\pm(x,y) \leq \operatorname{diam} \varphi_{s_n}(I_n) \to 0$. □

Lemma 3.14. *Suppose $X \subset I \times I$ is closed and $I \times \{0,1\} \subset X$. If no component Y of X satisfies $I \times \{0,1\} \subset Y$, then there is a path $\alpha : [0,1] \to I \times I \setminus X$ such that $\alpha(0) \in \{0\} \times I$ and $\alpha(1) \in \{1\} \times I$.*

Proof. For simplicity, we write $Q = I \times I$, $B = I \times \{0\}$, and $T = I \times \{1\}$. As X has at least two component, partition X into $X = X_T \sqcup X_B$ where $T \subset X_T$ and $B \subset X_B$ and $X_T \cap X_B = \varnothing$. Embed Q as the square in $\mathbb{R}^2$ with corners $(0,0), (0,1), (1,0), (1,1)$. We extend the sets X_T and X_B as $Y_T = X_T \cup \mathbb{R} \times [1,\infty)$ and $Y_B = X_B \cup \mathbb{R} \times (-\infty, 0]$; note that $Y_T \cap Y_B = \varnothing$ as well. Let U_T and U_B denote the complements of Y_T and Y_B respectively. We apply the Mayer-Vietoris sequence to $\mathbb{R}^2 = U_T \cup U_B$:

$$0 \to \tilde{H}_0(U_T \cap U_B) \xrightarrow{\psi} \tilde{H}_0(U_T) \oplus \tilde{H}_0(U_B) \to 0$$

is exact.

Pick points $p_L \in \{0\} \times I \setminus X$ and $p_R \in \{1\} \times I \setminus X$. As $U_B \setminus Q$ is path connected, we may join p_L and p_R with a path $\gamma_B \subset U_B$ which travels around and above Q—thus p_L and p_R are homologous as elements of $\tilde{H}_0(U_B)$. Likewise, we may connect them with a similar path $\gamma_T \subset U_T$ so they are also homologous in $\tilde{H}_0(U_T)$. Now consider $[p_L], [p_R] \in \tilde{H}_0(U_T \cap U_B)$; as $[p_L] = [p_R]$ in both $\tilde{H}_0(U_T)$ and $\tilde{H}_0(U_B)$, $\psi([p_L] - [p_R]) = 0$, but ψ is an isomorphism hence $[p_L]$ and $[p_R]$ are homologus in $\tilde{H}_0(U_T \cap U_B)$.

This produces an arc γ such that $\gamma(0) = p_L$, $\gamma(1) = p_R$ and $\gamma(t) \in U_T \cap U_B$ for all t. It may not be the case that $\gamma_t \in Q$ for all t, but some component of $\gamma(I) \cap Q$ must hit both the left and right edges—we take α to be this component. □

For the next two lemmas, we adopt the following conventions: We write E for the equator of S^2 and $\mathring{H}_\pm, \overline{H_\pm}$ for the open and closed northern and southern hemispheres respectively. In order to push around stems, we need a change of perspective; instead of the stem α being fixed while $\varphi_s(S^1)$ moves, we deal with a fixed circle (the equator) and an isotopy of arcs starting in the northern hemisphere.

Lemma 3.15. *Fix $x, y \in E \subset S^2$, points in the equator, and let γ_s, $s \in [0,1]$, be a continuous family of arcs in S^2 such that*

(1) For all s, $\gamma_s(0) = x$ and $\gamma_s(1) = y$.
(2) γ_0 is a proper arc in $\overline{H_+}$

Let $K = \{z \in E \mid z = \gamma_s(t), \text{ for some } s,t \in [0,1]\}$. Then there is a connected set $X \subset K \cup \big(\gamma_1(I) \cap \overline{H_+}\big)$ containing x and y.

Proof. We write $Q = I \times I$; define $\Gamma : Q \to S^2$ by $\Gamma(s,t) = \gamma_s(t)$. Let $A = \Gamma^{-1}(E)$ and $B = \Gamma^{-1}(\overline{H_+}) \cap \{1\} \times I$. If $p = (s,t) \in A$, then $\gamma_s(t) \in E$—this is the exact condition for $\gamma_s(t) \in K$—so $\Gamma(p) \in K$. Similarly, if $p \in B$, then $\Gamma(p) \in \gamma_1(I) \cap \overline{H_+}$ by construction.

Claim. There does not exist a path $\alpha : I \to Q \setminus (A \cup B)$ such that $\alpha(0) \in \{0\} \times I$ and $\alpha(1) \in \{1\} \times I$. ♮

Proof (of claim). Suppose otherwise. As $\alpha(I) \cap A = \varnothing$, $\Gamma \circ \alpha$ is a path in $S^2 \setminus E$. We know that $\alpha(0) \in \gamma_0 \setminus E$, hence $\alpha(0) \in \mathring{H}_+$. On the other hand, $\alpha(1) \in \gamma_1 \setminus H_+$, hence $\alpha(1) \in S^2 \setminus \mathring{H}_-$. But no arc can cross from $\mathring{H}_+$ to $\mathring{H}_-$ without intersecting E, a contradiction. □

As $I \times \{0\}, I \times \{1\} \subset A \cup B$, $A \cup B$ is closed, and no such arc α exists, by the contrapositive of Lemma 3.14, some component Y of $A \cup B$ contains both $I \times \{0\}$ and $I \times \{1\}$. We set $X = \Gamma(Y)$. We have seen that $\Gamma(A) \subset K$ and $\Gamma(B) \subset \gamma_1(I) \cap \overline{H_+}$, hence $X \subset K \cup \big(\gamma_1 \cap \overline{H_+}\big)$, as required. □

FIGURE – the isotopy of an arc in H_+, the square $I \times I$, and the connected sets we turn into the stems.

Lemma 3.16. *Let $\varphi_s : S^1 \to S^2$, $s \in [0,1]$, be an isotopy and $\alpha : I \to S^2$ be a proper path in $\overline{C_0^+}$ such that $\alpha(0) = \varphi_0(x)$ and $\alpha(1) = \varphi_0(y)$. Let $\beta_s = \varphi_{[0,s]}(x) * \alpha * \varphi_{[0,s]}(y)$, where $*$ is the path product and we reverse the initial or terminal path as necessary, and define $K = \{\varphi_1(z) \mid \varphi_s(z) = \beta_s(t) \text{ for some } s,t \in [0,1]\}$. There is a connected set $Y \subset K \cup (\alpha(I) \cap \overline{C_1^+})$ containing $\varphi_1(x)$ and $\varphi_1(y)$.*

Proof. Identifying S^1 with $E \subset S^2$ and using Theorem 2.15, extend φ_s to a family of maps $\Phi_s : S^2 \to S^2$. Let β_s be as above; then $\gamma_s = \Phi_s^{-1} \circ \beta_s$ is a isotopy of paths in S^2 with fixed endpoints and $\gamma_0((0,1)) \subset \mathring{H}_+$.

Let $K_\Phi = \{z \in E \mid z = \gamma_s(t) \text{ for some } s,t \in [0,1]\}$ and notice $K = \Phi_1(K_\Phi)$. Applying Lemma 3.15, we obtain a connected set $\tilde{Y} \subset K_\Phi \cup (\gamma_1(I) \cap H_+)$. We claim that $Y = \Phi_1(\tilde{Y})$ is the required set. By construction,

$$\begin{aligned} Y &\subset \Phi_1(K_\Phi) \cup \Phi_1(\gamma_1(I) \cap H_+) \\ &\subset \Phi_1(K_\Phi) \cup (\Phi_1 \Phi_1^{-1} \beta_1(I) \cap \Phi_1(H_+)) \\ &= K \cup (\beta_1(I) \cap \overline{C_1^+}). \end{aligned}$$

□

Corollary 3.17. *Suppose φ_s and α are as above and for all $s \in [0,1]$ $d_{\sup}(\varphi_s, \varphi_0) < \varepsilon$. Then* $\operatorname{diam} Y \leq \operatorname{diam} \alpha(I) + 4\varepsilon$.

Proof. Suppose $p \in Y$; we will show that $d_{S^2}(p, \alpha(I)) < 2\varepsilon$. As $d_{\sup}(\varphi_s, \varphi_0) < \varepsilon$, $d_{HD}(\beta_s(I), \alpha(I)) < \varepsilon$. Thus, $\operatorname{diam} \beta_s(I) \leq \operatorname{diam} \alpha(I) + 2\varepsilon$ and therefore if $p \in \beta_s(I)$ then $d_{S^2}(p, \alpha(I)) < 2\varepsilon$.

Now suppose that $p \in K$. By definition, $p = \varphi_1(z)$ and there are $s_0, t_0 \in [0,1]$ such that $\varphi_{s_0}(z) = \beta_{s_0}(t_0)$. But then

$$d_{S^2}(p, \alpha(I)) \leq d_{S^2}(p, \varphi_{s_0}(z)) + d_{S^2}(\varphi_{s_0}(z), \alpha(I))$$

The first term is less than ε by assumption, and the second is less that ε as $\varphi_{s_0}(z) \in \beta_{s_0}(I)$ and $d_{HD}(\beta_{s_0}(I), \alpha(I)) < \varepsilon$ as above. As $Y \subset K \cup \beta_1$, $d_{S^2}(p, \alpha(I)) < 2\varepsilon$ and the result follows. □

Proposition 3.18. *Suppose φ_s is a pinched Jordan curve and let $s_n \to s \in [0, \infty]$ be any convergent sequence. For any $x, y \in S^1$, $\lim_{n\to\infty} d^{\pm}_{s_n}(x,y)$ exists. If $s < \infty$, then* $\lim_{n\to\infty} d^{\pm}_{s_n}(x,y) = d_s(x,y)$.

Proof. To show that the limit exists, we will show that $d^{\pm}_{s_n}(x,y)$ is Cauchy. Fix $\varepsilon > 0$. As $\varphi_{s_n} \to \varphi_s$ uniformly, we may pass to a tail of the sequence and guarantee that $d_{\sup}(\varphi_{s_n}, \varphi_{s_m}) < \varepsilon$ for all m, n. Let X be an ε-efficient s_n stem and $\tilde{X}$ the s_m-stem obtained by applying Lemma 3.16 and Corollary 3.17 to X. Then,

$$d^{\pm}_{s_m}(x,y) \leq \operatorname{diam} \tilde{X} \leq \operatorname{diam} X + 4\varepsilon \leq d^{\pm}_{s_n}(x,y) + 5\varepsilon.$$

Likewise, $d^{\pm}_{s_n}(x,y) \leq d^{\pm}_{s_m}(x,y) + 5\varepsilon$ by a symmetric argument; whence $\lim_{s_n \to s} d^{\pm}_{s_n}(x,y)$ exists.

Now suppose that $s < \infty$ and X_n is a taut s_n-stem. Passing to a subsequence if necessary, we may assume $X_n \xrightarrow{HD} X$. Evidently, X is an s-stem so $d^{\pm}_s(x,y) \leq \lim_{n\to\infty} d^{\pm}_{s_n}(x,y)$. Now let Y be an ε-efficient s-stem. As we have passed to the tail of the sequence above, for any n, we may create Y', an s_n-stem such that $\operatorname{diam} Y' \leq \operatorname{diam} Y + 4\varepsilon$. But then,

$$d^{\pm}_{s_n}(x,y) \leq \operatorname{diam} Y' \leq \operatorname{diam} Y + 4\varepsilon \leq d^{\pm}_s(x,y) + 5\varepsilon.$$

After taking limits and letting $\varepsilon \to 0$, we conclude $\lim_{s_n} d^{\pm}_{s_n}(x,y) = d^{\pm}_s(x,y)$. □

Theorem 3.19. *Suppose that φ_s is a pinched Jordan curve. The inner distance*

$$\begin{aligned} d_\cdot(\cdot,\cdot) : S^1 \times S^1 \times [0,\infty] &\longrightarrow [0,\infty) \\ (x,y,s) &\longmapsto d_s(x,y) \end{aligned}$$

is continuous.

Proof. We define $d^{\pm}_\infty(x,y) = \lim_{s\to\infty} d^{\pm}_s(x,y)$ and by Proposition 3.18 this extension is well defined. Suppose $x_n \to x$ and $y_n \to y$ are convergent sequences in S^1 and $s_n \to s \in [0,\infty]$. Then

$$\begin{aligned} \left| d^{\pm}_s(x,y) - d^{\pm}_{s_n}(x_n,y_n) \right| &\leq \left| d^{\pm}_s(x,y) - \left(d^{\pm}_{s_n}(x_n,x) + d^{\pm}_{s_n}(x,y) + d^{\pm}_{s_n}(y_n,y) \right) \right| \\ &\leq \left| d^{\pm}_s(x,y) - d^{\pm}_{s_n}(x,y) \right| + d^{\pm}_{s_n}(x_n,x) + d^{\pm}_{s_n}(y_n,y). \end{aligned}$$

The first term goes to zero by Proposition 3.18 and the latter terms by Lemma 3.13. □

4. The Inner Decompositions of a Pinched Jordan Curve

In this section, we show that associated to every surjective pinched Jordan curve there is a pair of transverse singular foliations of the disk $\mathcal{D}$ extending the decompositions $\Lambda^{\pm}$. The following are definitions from [FL25]:

Definition 4.1. A decomposition Λ of a circle S is *special* if it is upper semi-continuous and has the following properties:

(1) *unlinking*: distinct $\lambda, \lambda' \in \Lambda$ are unlinked
(2) *nesting*: given any $\lambda \in \Lambda$ and a disjoint compact interval $I \subset S$, there is a decomposition element $\lambda' \in \Lambda$ which separates I from λ.

A pair of decompositioins $\Lambda^{\pm}$ of S have *efficient intersection* if $\lambda^+ \cap \lambda^-$ is at most one point for any $\lambda^+ \in \Lambda^+$ and $\lambda^- \in \Lambda^-$.

We call a pair of special decompositions $\Lambda^{\pm}$ of S an *especial pair* if they have the efficient intersection property. ♮

Definition 4.2. Let D denote a closed disk and let S be its boundary. A decomposition $\mathcal{S}$ of D is *unbounded* if each $K \in \mathcal{S}$ intersects S and it is *monotone* if each of its elements are connected.

A pair of decompositions $\mathcal{S}^{\pm}$ have *efficient intersection* if $K^+ \cap K^-$ is at most one point for each $K \in \mathcal{S}^+$ and $K^- \in \mathcal{S}^-$.

A pair of monotone, upper semi-continuous, unbounded decompositions $\mathcal{S}^{\pm}$ of D are called an *muu* pair. If the pair also has efficient intersection, then they form an *emuu* pair. ♮

Theorem 4.3 ([FL25], Theorem 3.4)**.** *Let $\Lambda^{\pm}$ be an especial pair on a circle S. Then there are decompositions $\mathcal{F}^{\pm}$ of a disk $\mathcal{D} = D \sqcup S$ with the following properties:*

(1) $\mathcal{F}^{\pm}$ are an emuu pair with $\partial\mathcal{F}^{\pm} = \Lambda^{\pm}$ and is the unique pair up to a homeomorphism of $\mathcal{D}$ that fixes S.
(2) The decompositions $\mathcal{F}^{\pm}$ determine transverse pair of essential foliations of D given by

$$F^{\pm} = \{\textit{Components of } K \cap D \mid K \in \mathcal{F}^{\pm}\}$$

Theorem 4.11 follows from showing that for pinched Jordan curves with no left or right side, the pinched pseudometrics d_{∞} determine an especial pair $\Lambda^{\pm}$.

Definition 4.4. Let φ be a pinched Jordan curve and $d_{\infty}^{+}\pm$ corrosponding the pinched pseudometrics. Define the *left inner decomposition* $\sim_+$ (resp. *right inner decomposition* $\sim_-$) on S^1 by $x \sim_+ y$ if $d_{\infty}^{+}(x, y) = 0$ (resp. $x \sim_- y$ if $d_{\infty}^{-}(x, y) = 0$). Let $\Lambda^{\pm}$ denote the set of equivalence classes of $\sim_{\pm}$ ♮

Verifying that $\sim_{\pm}$ determine an equivalence follows directly from the definition of $d_{\infty}^{\pm}$.

Lemma 4.5. *Every $\lambda \in \Lambda^{\pm}$ is closed as a subset of S^1.*

Proof. Let $\lambda \in \Lambda^+$ and let x_n be a sequence of points in λ converging to some point $x \in S^1$. For all n, $d_{\infty}^{+}(x_1, x_n) = 0$. As d_{∞}^{+} is continuous, it follows that $d_{\infty}^{+}(x_1, x) = 0$ as well. □

The following is essentially [Cal07, Lemma 10.13].

Lemma 4.6. *Let λ and λ' be distinct equivalence classes in Λ^+. Then λ and λ' are unlinked as subsets of S^1*

Proof. Assume that $\lambda, \lambda' \in \Lambda^+$ link in S^1. In particular, we can find $x, y \in \lambda$ and $z, w \in \lambda'$ such that $\{x, y\}$ and $\{z, w\}$ link. Let $s_n \to \infty$ and suppose X_n are taut s_n-stems for x and y and Z_n are taut s_n-stems for z and w. As $\{x, y\}$ and $\{z, w\}$ link, it must be the case that $X_n \cap Z_n \neq \varnothing$ for all n. But then $X_n \cup Z_n$ is an s_n-stem for x and z and $\operatorname{diam}(X_n \cup Z_n) \leq \operatorname{diam} X_n + \operatorname{diam} Z_n$. Thus, $d^+_\infty(x, z) = 0$ and therefore $\lambda = \lambda'$. □

Lemma 4.7. *If $x, y \in S^1$ are distinct, then at most one of $d^\pm_\infty(x, y) = 0$.*

Proof. Suppose to the contrary. Fix sequences $\varepsilon_k \to 0$ and $s_k \to \infty$ and let α_k and β_k denote ε_k-efficient stems in $\overline{C_{s_k^+}}$ and $\overline{C_{s_k^-}}$ respectively. Then $J_k = \alpha_k \cup \beta_k$ is a Jordan curve. If U is any neighborhood of $p = \varphi_\infty(x)$, we may throw away finitely many terms and assume $J_k \subset U$. Let V_k denote the component $S^2 \setminus J_k$ contained in U and I_k the component of $S^1 \setminus \{x, y\}$ such that $\varphi_k(I_k) \subset V_k$. After passing to the tail of the sequence, we may assume that I_k is the same component for all k—so we write I for I_k. We know that $\varphi_k(I) \subset V_k$ and hence $\varphi_\infty(I) \subset \{p\}$. As φ_∞ must have totally disconnected preimages, we have arrived at a contradiction. □

Lemma 4.8. *The decompositions $\Lambda^\pm$ are upper semi-continuous.*

Proof. Let $K \subset S^1$ be closed. We will show that $L = \bigcup\{D \subset \mathcal{D} \mid D \subset U\}$ is closed—it will then follow, by Proposition 2.9 (2), that Λ^+ is then upper semi-continuous. Suppose that x_k is a sequence in L converging so some $x \in S^1$. Associated to each x_k is a leaf λ_k. After passing to a subsequence if necessary, assume that $\lambda_k \xrightarrow{HD} \lambda$. By construction, there $y_k \in \lambda_k \cap K$; as K is compact we may pass to another subsequence and assume that $y_k \to y \in K$. As $x_k, y_k \in \lambda_k$, $d^+_\infty(x_k, y_k)$ for all k. But d^+_∞ is continuous so $d^+_\infty(x, y) = 0$ as well. Thus, $x \in L$ and L is closed. □

Let X denote $\mathbb{R}^2$ or S^2. By a *rectangle* in X we mean an embedding of $[0, 1] \times [0, 1]$ in X. We say that a closed disk $D \subset X$ is μ-thin when $\mu = \sup\{r \mid B_r(p) \subset D \text{ for some } p \in D\}$.

Lemma 4.9. *Suppose $Q \subset \mathbb{R}^2$ is a rectangle and let L, R, T, B denote the left, right, top, and bottom edged of the rectangle respectivley. Suppose further that there are constants δ, μ such that*

(1) $d_Q(L, R) > \delta$, where d_Q is inner distance on Q.
(2) Q is μ-thin

If $\mu < \delta/2$, then there are points $p \in T$, $q \in B$ such that $d_Q(p, q) < 2\mu$, $d_Q(p, L \cup R) > \delta/2$, and $d_Q(q, L \cup R) > \delta/2$.

Proof. In this proof, we use the d_Q metric to create our neighborhoods. As $d_Q(L, R) > \delta$, $N_{\delta/2}(L) \cap N_{\delta/2}(R) = \varnothing$. Let γ be any path in $Q \setminus (N_{\delta/2}(L) \cup N_{\delta/2}(R))$ starting in $T \setminus (N_{\delta/2}(L) \cup N_{\delta/2}(R))$ and ending in $B \setminus (N_{\delta/2}(L) \cup N_{\delta/2}(R))$. As Q is μ-thin, $\{N_\mu(L), N_\mu(R), N_\mu(T), N_\mu(B)\}$ covers Q, but as $\mu < \delta/2$ so does $\{N_{\delta/2}(L), N_{\delta/2}(R), N_\mu(T), N_\mu(B)\}$. As $\gamma \subset Q \setminus (N_{\delta/2}(L) \cup N_{\delta/2}(R))$, $\{N_\mu(T), N_\mu(B)\}$ covers γ. It follows that $N_\mu(T) \cap N_\mu(B) \cap \gamma$ is nonempty—indeed $N_\mu(T) \cap \gamma$ and $N_\mu(B) \cap \gamma$ are open subsets of γ covering γ, hence they must intersect as γ is connected.

Thus, there is some point $x \in \gamma \cap N_\mu(T) \cap N_\mu(B)$. By construction, there are point $p \in T$, $q \in B$ such that $d_Q(x, p), d_Q(x, q) \leq \mu$—thus $d_Q(p, q) \leq 2\mu$, as required. □

Proposition 4.10. *Let φ be a pinched Jordan Curve with no left side. Then Λ^+ satisfies the nesting condition.*

Proof. Select $\lambda \in \Lambda^+$, $U = (a, b)$ a complementary region to λ, and let $I = [x_1, x_2] \subset U$ be a closed interval. Since I is contained in a complementary interval to λ, the images $\varphi(I)$ and $\varphi(\lambda)$ must be disjoint. Let $\delta = \inf_{x \in I} d^+_\infty(a, x)$; as d^+_∞ is continuous, $\delta > 0$.

We now fix some $x \in I$ and choose a sequence $s_n \to \infty$. For each n, let α_n be a $1/n$-efficient s_n-stem for x and a. Observe that for each n, there is a rectangle $Q_n \subset \overline{C^+_{s_n}}$ with sides α_n, $\varphi_{s_n}(I)$, and $\varphi_{s_n}(U \setminus I)$—denote these as L_n, R_n, T_n, B_n respectively. First, we will find points z, w in different comopnents of $U \setminus I$, then we will show that (z, w) is a leaf in Λ^+.

For each n, let ε_n denote the thinnness of $\overline{C^+_{s_n}}$; note that $\varepsilon_n \to 0$. As φ_∞ has no left side, if n is sufficiently large, then we may assume that $\varepsilon_n < \delta/4$ and $d_{s_n}(L_n, R_n) > \delta/2$. These are the exact hypotheses of Lemma 4.9, so there are points $p_n \in T_n$ and $q_n \in B_n$ such that $d_{Q_n}(p_n, q_n) < 2\varepsilon_n$. Now let $z_n, w_n \in U \setminus I$ be sequences such that $\varphi_{s_n}(z_n) = p_n$ and $\varphi_{s_n}(w_n) = q_n$. Passing to a subsequence if necessary, assume that $z_n \to z, w_n \to w$. As every p_n and q_n are at least $\delta/4$ away from L_n and R_n, $p, q \in U \setminus I$ as well.

Then

$$d_{s_n}(z, w) \leq d_{s_n}(z, z_n) + d_{s_n}(z_n, w_n) + d_{s_n}(w_n, w)$$

The outer terms tend to zero by Lemma 3.13 and the central term is bounded by $2\varepsilon_n$ by construction. Taking limits, we have some leaf λ' containing z and w. As z and w lie in different components of $U \setminus I$, λ' separates λ and I. □

Decompositions of S^1 with the nesting property have a rigid structure. Suppose that $\{x, y\} \in \Lambda^+$ contains exactly two points and write $S^1 \setminus \{x, y\} = I \sqcup J$. There are sequences of leaves $\lambda_n \subset I$ and $\mu_n \subset J$ such that $\lambda_n \xrightarrow{HD} \{x, y\}$ and $\mu_n \xrightarrow{HD} \{x, y\}$. On the other hand, if $\lambda \in \Lambda^+$ contains more than two points, then it determines a (possibly infinite sided) polygon inscribed in S^1 and every side of this polygon can be realized as a limit of leaves contained in the corrosponding component of $S^1 \setminus \lambda$.

Theorem 4.11. *Let φ be a surjective pinched Jordan curve and $\mathcal{D} = D \sqcup S^1$. Associated to φ is a pair of laminations $\Lambda^\pm$ which extend to pair of transverse singular foliations $F^\pm$ of D extending $\Lambda^\pm$.*

Proof. By Lemmas 4.5 to 4.8 and proposition 4.10, $\Lambda^\pm$ form an especial pair in S^1. Then Theorem 4.3 produces a pair of decompositions $\mathcal{F}^\pm$ of $\mathcal{D}$ extending $\Lambda^\pm$ and $F^\pm = \{\text{Components of } K \cap D \mid K \in \mathcal{F}^\pm\}$ produces the required singular foliations. □

5. Special Decompositions are Dendrites

In this section we study the decomposition spaces $\Lambda^\pm$ of a pinched Jordan curve. In particular, we show that they are each homeomorphic to a compact *real tree*, a uniquely arcwise connected and locally arcwise connected space; the prototypical example of a real tree is an $\mathbb{R}$-tree considered without its metric. On the other hand, if real tree T is metrizable, by [MO90] and [Bow99] T can be given a metric where it is an $\mathbb{R}$-tree. A *dendrite* is a compact, separable real tree. Given any separable $\mathbb{R}$-tree T, we may give T the weak *observer's topology* and then compactify T into a dendrite [CHL07].

Definition 5.1. A decomposition $\mathcal{D}$ of a space X is said to be *nonseparated* if, given any $D \in \mathcal{D}$, no $D' \in D$ lies in more than one component of $X \setminus D$. ♮

Definition 5.2. A decomposition $\mathcal{D}$ of a space X is said to be *saturated* if, given any $D \in \mathcal{D}$ and $x \in X \setminus D$, there is some $D' \in \mathcal{D}$ separating x and D. ♮

In [Why63], it is first shown that any nonseparated saturated decomposition of a compact space is automatically upper semicontinuous (VII.4 Thm. 4.21). If the initial space is a continuum, there is more to say.

Proposition 5.3 ([Why63], VII.4 Thm. 4.23)**.** *Let $\mathcal{D}$ be a nonseparated and saturated decomposition of a continuum X. Then the decomposition space $\mathcal{D}$ is a dendrite.*

We will show that $\Lambda^{\pm}$ are nonseparated and saturated. To do so, we will work both $\Lambda^{\pm}$ and a corresponding muu decomposition $\mathcal{F}^{\pm}$ of $\boldsymbol{\mathcal{D}} = D \sqcup S^1$ extending $\Lambda^{\pm}$. We justify this with the following lemma.

Lemma 5.4. *Let Λ be a special decomposition of S^1 and $\mathcal{F}$ a muu decomposition which extends Λ. As decomposition spaces, Λ and $\mathcal{F}$ are homeomorphic*

Proof. Consider the inclusion $S^1 \hookrightarrow \boldsymbol{\mathcal{D}}$. Since $\mathcal{F}$ extends Λ, the inclusion respects the decompositions Λ and $\mathcal{F}$ and therefore descends to a continuous bijection between Λ and $\mathcal{F}$. Λ is compact and by Proposition 2.9 $\mathcal{F}$ is Hausdorff, hence they are homeomorphic. □

We state the theorem below for a special decomposition of the circle, but we note that the proof automatically extends the result to any muu decomposition of a closed disk as well.

Theorem 5.5. *Let Λ be a special decomposition of S^1. Then the decomposition space Λ is a dendrite.*

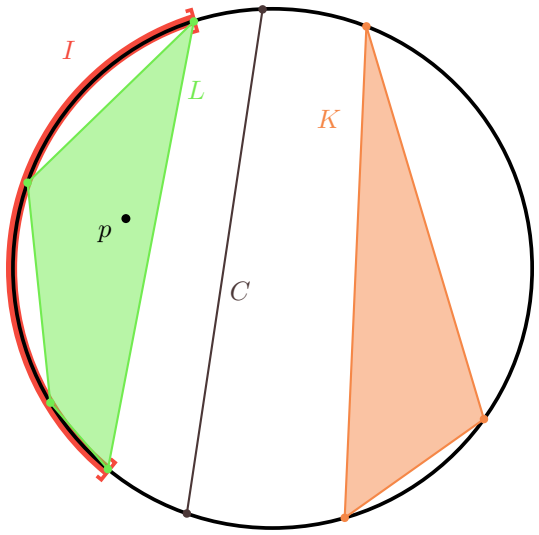


FIGURE 1. TODO

Proof. By Lemma 5.4, we may use any muu decomposition $\mathcal{F}$ that extends Λ, so we will assume that $\mathcal{F}$ is the hull decomposition $\mathcal{F} = H(\Lambda)$. It is clear that monotone decomposition are nonseparated, so we must establish that $\mathcal{F}$ is saturated. Fix $K \in \mathcal{F}$ and $x \in \boldsymbol{\mathcal{D}} \setminus K$. Let $L = L(x) \in \mathcal{F}$ denote the unique decomposition element containing x, L_Λ the restriction of L to Λ and K_Λ the restriction of K to Λ. Explicitly, L_Λ and K_Λ are given by intersection L and K with S^1. As distinct elements of Λ are unlinked, some complementary region U of $S^1 \setminus K_\Lambda$ contains all of L_Λ. Let $I \subset U$ denote the convex hull of L_Λ. Then I is a closed

interval in the complement of K_Λ, so nesting implies there is a some $C \in \Lambda$ separating K_Λ and I—hence separating K_Λ and L_Λ. As $\mathcal{F}$ is the hull decomposition, it follows that $H(C)$ separates K and L. Thus, $\mathcal{F}$ is a dendrite by Proposition 5.3 and by Lemma 5.4 so is Λ. □

6. Self Bumping and Laminations

In [CL26], Calegari and Loukidou discus the notion of a "self-bumping" $\varphi \in \operatorname{Fr}\operatorname{Emb}(S^1, S^2)$. Such a φ can be approximated by embeddings in more than one way—the pinched laminations give a partial answer to untangling the ambiguity of self-bumping points.

We say that two maps $f, g : X \to Y$ between metric spaces are *δ-isotopic* if there is an isotopy $F : X \times I \to Y$ between f and g such that for all $x \in X$, $\operatorname{diam}\{x \times I\} < \delta$. Define an equivalence relation on the set of pseudo-isotopies landing on a fixed $\xi \in \operatorname{Fr}\operatorname{Emb}(S^1, S^2)$ as follows: $\varphi \sim \psi$ if for every $\varepsilon > 0$ there is a $\delta > 0$ with $\delta \to 0$ as $\varepsilon \to 0$ such that if for all $x \in S^1$, $d(\varphi_s(x), \xi(x)) < \varepsilon$ and $d(\psi_s(x), \xi(x)) < \varepsilon$ then φ_s and ψ_s are δ-isotopic. The goal of this final section is to prove the following:

Theorem 6.1 (Calegari-Kerbs)**.** *For a fixed $\xi \in \operatorname{Fr}\operatorname{Emb}(S^1, S^2)$ and two pseudo-isotopies Φ and Ψ landing on ξ, $\Phi \sim \Psi$ if and only if the pinched laminations agree.*

Appendix A. Cannon-Thurston Maps and Pseudo-Isotopies

In this section we show that the Cannon-Thurston maps arising from surface bundles [CT07] and from quasigeodesic flows [Fra15] are approximated by pseudo-isotopy. As a consequence, the corrosponding singular foliations agree with those constructed in Theorem 4.11.

A.1. Shrinkable Decompositions.

Definition A.1. Let X be a compact metric space, $\mathcal{D}$ an upper semi-continuous decomposition of X, and ρ any metric on $\mathcal{D}$. We say $\mathcal{D}$ is *shrinkable* if, for every $\varepsilon > 0$ there exists a homeomorphism $h : X \to \mathcal{D}$ such that:

(1) $\rho(\pi h(x), \pi(x)) < \varepsilon$ for all $x \in X$
(2) $\operatorname{diam} h(D) < \varepsilon$ for all $D \in \mathcal{D}$. ♮

One can make sense of shrinkable decomposition in the non-metric case. The reformulation involves comparing arbitrary open covers to arbitrary saturated open covers. The definition says that elements of $\mathcal{D}$ can be made small without moving points too far relative to $\mathcal{D}$. This can be upgraded with a routine application of the Lebesgue covering lemma.

Proposition A.2 ([Dav], II.5 Thm. 2)**.** *Let $\mathcal{D}$ be an upper semi-continuous decomposition of a compact metric space X. The decomposition map $\pi : X \to \mathcal{D}$ can be approximated by homeomorphisms if and only if $\mathcal{D}$ is shrinkable.*

Remark A.3. Shrinkable decompositions preserve local topological properties. Suppose that that X is a compact metric space and $\mathcal{P}$ a topological property which applies to subsets of a space. Suppose that every $x \in X$ has arbitrarily small neighborhoods with property $\mathcal{P}$. Then every element D of the decomposition space $\mathcal{D}$ has arbitrarily small neighborhoods with property $\mathcal{P}$. ♮

Given a compact set $K \subset X$, we say that K is *cell-like* if, for every neighborhood U of K, the inclusion $\imath : K \hookrightarrow U$ is nullhomotopic. In the case where X is a 2-manifold, every cell-like set is cellular—that is, it can be written as the intersection of countably many nested closed balls. Every cellular set is evidently cell-like, but the converse is false—counterexamples can be found in [Dav] or [Kal21]. A decomposition $\mathcal{D}$ is said to be cell-like if every element is cell-like.

Lemma A.4. *Every compact, contractable set K in a manifold X is cell-like.*

Lemma A.5 ([Kal21] Thm. 5.4, (1))**.** *Let $\mathcal{D}$ be an upper semi-continuous cell-like decomposition of a surface X. Then $\mathcal{D}$ is shrinkable.*

In the context of Proposition A.2, one might wonder if the sequence of homeomorphisms approximating π can be interpolated to a pseudo-isotopy.

Definition A.6. If X has an upper semi-continuous decomposition $\mathcal{D}$ such that $\mathcal{D} = \{\varphi_\infty^{-1}(y) \mid y \in Y\} \setminus \varnothing$, then we say that $\mathcal{D}$ is *realized by pseudo-isotopy.* ♮

The exact decompositions which admit realization by pseudo-isotopy have been well studied. For our work, the following suffices.

Proposition A.7 ([Dav], II.13 Thm. 4)**.** *If $\mathcal{D}$ is a shrinkable decomposition of a manifold M then $\mathcal{D}$ can be realized by a pseudo-isotopy. Further, if $\mathcal{U}$ is a $\mathcal{D}$-saturated open cover of M, then there exists a pseudo-isotopy $\varphi : M \times I \to M$ such that for each $p \in M$, $\varphi(\{p\} \times I) \subset U$ for some $U \in \mathcal{U}$.*

A.2. **Cannon-Thurston maps are realized by pseudo-isotopy.** Given a quasigeodesic flow Ψ on a closed hyperbolic 3-manifold M, Frankel constructs a $\pi_1 M$ equivariant Peano curve $S^1 \to S^2$ [Fra15]. Here, the S^1 is the compactified boundary of the flowspace of Ψ and the S^2 is the sphere at infinity of $\tilde{M}$. This generalizes the earlier work of Cannon and Thurston [CT07] who construct a Peano curve in the case where M fibers over the circle. These Peano curves are all realizable by pseudo-isotopy. By Proposition A.7, it is sufficient to construct a shrinkable decomposition such that the decomposition map realizes the Peano curve.

If M fibers over the circle, then M comes equipped with a suspension pseudo-Anosov flow Φ given by the monodromy. Let $\boldsymbol{\mathcal{P}} = \mathcal{P} \sqcup S^1_u$ denote the compactified flowspace of Φ (which we may view as the universal cover of some fiber Σ). Glue two copies of $\boldsymbol{\mathcal{P}}$ along S^1_u to form the *universal sphere*, S^2_u. $\boldsymbol{\mathcal{P}}$ supports a pair of tranverse singular foliations—each of which is a monotone, upper semi-continuous decomposition. We may extend these foliations to a decomposition $\mathcal{D}_\Phi$ of S^2_u by foliating the northern and southern hemisphere with the positive and negative foliations respectively. We stress that S^2_u is not foliated—points in S^1_u generally lack foliated charts. As no positive leaf shares an end with a negative leaf, no leaf of $\mathcal{D}$ can separate S^2_u.

Notice that there is a natural map $\pi : S^2_u \to S^2_\infty$ given by the Cannon-Thurston map on S^1_u and collapsing each decomposition element in the hemispheres to the image of their corresponding point in S^2_∞. Then $\mathcal{D}_\Phi = \{\pi^{-1}(p) \mid p \in S^2_\infty\}$, so $\mathcal{D}$ is upper semicontinuous. As each decomposition element is cellular (indeed, it is either an arc, a finite pronged singular arc, or a point), $\mathcal{D}$ is cellular, hence by Lemma A.5 $\mathcal{D}$ is shrinkable and therefore realized by a pseudo-isotopy.

The case where Ψ is a quasigeodesic flow on a closed hyperbolic manifolds is similar. The flowspace $\boldsymbol{\mathcal{P}} = \mathcal{P} \sqcup S^1_u$ comes equipped with a pair of decompositions $\mathcal{D}^\pm$. We may construct the universal sphere the same way: $S^2_u = \boldsymbol{\mathcal{P}} \cup_{S^1_u} \boldsymbol{\mathcal{P}}$ and we can use $\mathcal{D}^\pm$ to generate a decomposition of S^2_u.

There are natural maps $e^\pm : \mathcal{P} \to S^2_\infty$ sending a flowline α to its forward or backward endpoint. In [Fra15], Frankel shows that that $e^\pm$ have natural extensions to S^1_u. Consider the map $m : S^2_u \to S^2_\infty$ which is e^+ on the (closed) northern hemisphere and e^- on the southern. As e^+ and e^- agree on S^1_u, m is well defined. In [Fra17, Section 5], Frankel uses m to construct the *master decomposition* of S^2_u:

$$\mathcal{M} = \{m^{-1}(p) \mid p \in S^2_\infty\},$$

a monotone and upper semi-continuous decomposition of S^2_u. Notice that this is the decomposition given by taking the transitive closure of $\mathcal{D}^{\pm}$ viewed as partial decomposition of S^2_u. He then shows that each $K \in \mathcal{M}$ is nonseparating—meaning $\mathcal{M}$ is a cell-like decomposition by Lemma A.4 and therefore shrinkable and realized by a pseudo-isotopy.

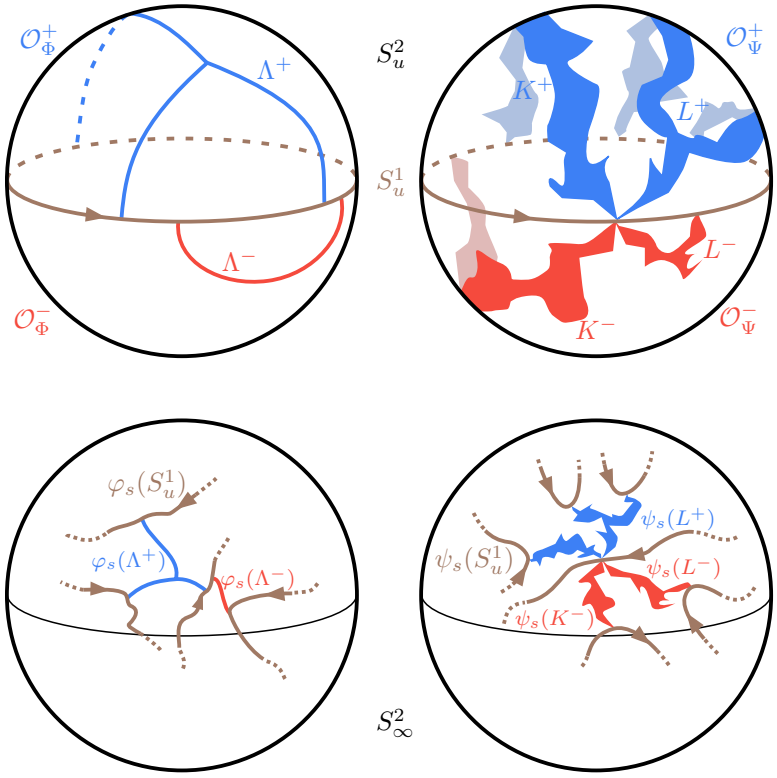


FIGURE 2. Top: The universal sphere associated a pseudo-Anosov suspension flow Φ and a quasigeodesic flow Ψ and some positive an negative decomposition elements. For a suspension flow, positive and negative decomposition elements do not meet at S^1_u, but this can happen for a quasigeodesic flow. Bottom: The image of the decomposition elements under a pseudo-isotopy realizing the decomposition.

It is worth commenting on the pseudo-isotopy produced by Proposition A.7. We construct it by interpolating between a sequence of homeomorphisms $S^2_u \to S^2_\infty$ which shrink decomposition elements to have diameter less than some $\varepsilon_n \to 0$—see Figure 2. In both cases, the intersection of each $D \in \mathcal{D}$ with its respective S^1_u is totally disconnected, it is not shrunk and instead winds its away around S^2_∞. In the limit, of course, the image of S^1_u is all of S^2_∞, realizing the Cannon-Thurston maps.

Consider one decomposition element $K \in \mathcal{M}$ and write $\mathcal{K} = \mathcal{K}^+ \cup \mathcal{K}^-$ for its intersections with $\mathcal{D}^{\pm}$—note that if both $\mathcal{K}^+$ and $\mathcal{K}^-$ are nonempty, then $\mathcal{K}^+ \cap \mathcal{K}^- \subset S^1_u$. Let K^+ be a component of $\mathcal{K}^+$ and K^- a component of $\mathcal{K}^-$. Write $\{x_n\} = K^+ \cap S^1_u$ and $\{y_n\} = K^- \cap S^1_u$. Note that $\pi(\{x_n\}) = \pi(\{y_n\}) = p$, but it is not necessarily the case that $\pi^{-1}(p) = \{x_n\} \cup \{y_n\}$. Under the pseudo-isotopy ψ_s, K^+ becomes a closed, connected set contained entirely to the

left of the Jordan curve $\psi_s(S^1_u)$ except at $\psi_s(\{x_n\})$, where it intersects $\psi_s(S^1_u)$—and the same can be said for $\psi_s(K^-)$'s containment to the right of $\psi_s(S^1_u)$.

References

[Bow99] Brian H. Bowditch. *Treelike Structures Arising from Continua and Convergence Groups*. Online-Ausg. American Mathematical Society volume 139, number 662. Providence, R.I: American Mathematical Society, 1999.

[Cal07] Danny Calegari. *Foliations and the Geometry of 3-Manifolds*. 1. publ. Oxford Mathematical Monographs. Oxford: Oxford Univ. Press, 2007.

[CHL07] Thierry Coulbois, Arnaud Hilion, and Martin Lustig. *Non-Unique Ergodicity, Observers' Topology and the Dual Algebraic Lamination for $\backslash R$-Trees*. June 2007. DOI: 10.48550/arXiv.0706.1313. arXiv: 0706.1313 [math].

[CL26] Danny Calegari and Ino Loukidou. *CaTherine Wheels*. https://arxiv.org/abs/2604.24619v1. Apr. 2026.

[CT07] James W Cannon and William P Thurston. "Group Invariant Peano Curves". *Geometry & Topology* 11.3 (July 2007), pp. 1315–1355. DOI: 10.2140/gt.2007.11.1315.

[Dav] Robert J. Daverman. *Decompositions of Manifolds*. American Mathematical Soc.

[EK71] Robert D. Edwards and Robion C. Kirby. "Deformations of Spaces of Imbeddings". *Annals of Mathematics* 93.1 (1971), pp. 63–88. DOI: 10.2307/1970753. JSTOR: 1970753.

[Fab99] Paul Fabel. ""Shortest" Arcs in Closed Planar Disks Vary Continuously with the Boundary". *Topology and its Applications* 95.1 (June 1999), pp. 75–83. DOI: 10.1016/S0166-8641(97)00275-7.

[FL25] Steven Frankel and Michael Landry. *From Quasigeodesic to Pseudo-Anosov Flows*. Oct. 2025. DOI: 10.48550/arXiv.2510.02217. arXiv: 2510.02217 [math].

[Fra15] Steven Frankel. "Quasigeodesic Flows and Sphere-Filling Curves". *Geometry & Topology* 19.3 (May 2015), pp. 1249–1262. DOI: 10.2140/gt.2015.19.1249. arXiv: 1210.7050 [math].

[Fra17] Steven Frankel. *Coarse Hyperbolicity and Closed Orbits for Quasigeodesic Flows*. Nov. 2017. DOI: 10.48550/arXiv.1507.04320. arXiv: 1507.04320.

[HY88] John G. Hocking and Gail S. Young. *Topology*. New York: Dover Publications, 1988.

[Kal21] Boldizsar Kalmar. *Decomposition Space Theory*. Mar. 2021. DOI: 10.48550/arXiv.2103.02977. arXiv: 2103.02977 [math].

[MO90] John C. Mayer and Lex G. Oversteegen. "A Topological Characterization of R-Trees". *Transactions of the American Mathematical Society* 320.1 (July 1990), p. 395. DOI: 10.2307/2001765. JSTOR: 2001765.

[OT08] Lex G. Oversteegen and E. D. Tymchatyn. *Extending Isotopies of Planar Continua*. Nov. 2008. DOI: 10.48550/arXiv.0811.0364. arXiv: 0811.0364 [math].

[Why63] Gordon Thomas Whyburn. *Analytic Topology*. American Mathematical Society, 1963.

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